

Self-Organizing Aerial Swarm Robotics for Resilient Load Transportation : A Table-Mechanics-Inspired Approach

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In comparison with existing approaches, which struggle with scalability, communication dependency, and robustness against dynamic failures, cooperative aerial transportation via robot swarms holds transformative potential for logistics and disaster response. Here, we present a physics-inspired cooperative transportation approach for flying robot swarms that imitates the dissipative mechanics of table-leg load distribution. By developing a decentralized dissipative force model, our approach enables autonomous formation stabilization and adaptive load allocation without the requirement of explicit communication. Based on local neighbor robots and the suspended payload, each robot dynamically adjusts its position. This is similar to energy-dissipating table leg reactions. The stability of the resultant control system is rigorously proved. Simulations demonstrate that the tracking errors of the proposed approach are 20%, 68%, 55.5%, and 21.9% of existing approaches under the cases of capability variation, cable uncertainty, limited vision, and payload variation, respectively. In real-world experiments with

six flying robots, the cooperative aerial transportation system achieved a 94% success rate under single-robot failure, disconnection events, 25% payload variation, and 40% cable length uncertainty, demonstrating strong robustness under outdoor winds up to Beaufort scale 4. Overall, this physics-inspired approach bridges swarm intelligence and mechanical stability principles, offering a scalable framework for heterogeneous aerial systems to collectively handle complex transportation tasks in communication-constrained environments.

Introduction

Load transport is a fundamental component of modern society. With the advent of diverse aerial vehicles (1–7), aerial transportation has emerged as a versatile solution for the transportation of cargo, transcending the constraints of geographical limitations and overcoming the limitations of ground and water transportation. However, in contrast to ground robots, which benefit from the capacity to transport heavy payloads over extended distances, aerial robots are constrained by thrust limitations. This has imposed strong limitations on their potential applications (8). Given the considerable expense and technical difficulties imposed on the development of large-scale flying robots, the use of multiple regular-sized flying robots is being considered as a potential solution for efficient transportation (9, 10). Several approaches to the cooperative transportation of flying robots are currently being explored (11). The rigid attachment has been a commonly used solution for addressing bar-shaped loads with two flying robots (12–15). The flying robots themselves can adjust the flight attitude of the load, but this introduces additional interference, which jeopardizes the stability of the robots. In parallel, soft attachment with cable addresses the load transportation, and the solution is insensitive to the flying attitude of robots. Furthermore, the low cost and convenient deployment of this soft attachment solution have led to increasing interest in working on cooperative transportation with flying robots (16–19). However, cooperative transportation continues to face significant difficulties. Unexpected communication interference poses a major obstacle to achieving precise formation control, while complex weather conditions add further uncertainty when operating in dynamic environments. Furthermore, mid-flight failures may occur, potentially jeopardizing flight safety (20). These practical scenarios and potential issues

drive a pressing need to develop a robust and generalized transportation framework to address these difficulties and ensure the safety of cooperative transport operations.

Why Cooperative Transportation of Cable Suspended Payloads with Aerial Robotics Intuitively, there are two primary approaches to increasing the payload capacity of transportation robots. The first is to design a larger aerial robot capable of carrying heavier loads, while the second involves coordinating multiple smaller aerial robots to work collaboratively for transport (21, 22). Although system complexity rises with the number of robots, the latter approach offers several significant advantages over the former and exhibits the following key features.

(i) **Scalable.** The payload capacity can be readily scaled by increasing the number of aerial robots. This cooperative strategy is commonly observed in nature, where individuals work together to transport food or nesting materials in a flexible manner, free from the constraints of a fixed number, formation, or group composition (23). Individuals can join or leave a transport mission without requiring the payload to be reallocated, significantly enhancing transportation efficiency. Moreover, this approach greatly reduces costs associated with design, testing, production, and maintenance (24). Additionally, the design process for flying robots with varying sizes and payload capacities can be streamlined, eliminating the need for repeated design iterations for different transportation tasks. Compared to a single large robot carrying heavy objects, a group of lightweight, small robots working together to transport heavy loads is more consistent with natural phenomena and evolutionary trends.

(ii) **Robust.** In a cooperative transportation mechanism, attaching multiple components to the payload enhances both the vehicle's stability and its control (19). The deployment of multiple flying robots inherently introduces redundancy, significantly improving the transportation process's overall safety and reliability. In the event of multiple robot failures, it is crucial that the remaining robots adapt to the sudden changes and continue to perform their mission. Compared to a single-robot scenario, a cooperative strategy enables a more robust and resilient transport process (22, 25), and the over-actuated nature of the cooperative transportation system allows multiple robots to stably control the suspended load in terms of both position and attitude, ensuring enhanced stability and safety throughout the operation.

(iii) **Flexible.** When transporting loads outdoors, complex environments such as narrow terrains

often require deformable formations to ensure safe passage. A cooperative transportation mechanism is well-suited to meet such demands due to the flexibility in both the number of robots and their formation. Additionally, cooperative strategies enable the implementation of various flying modes to accommodate different tasks. In scenarios where the load is deformable or has an irregular shape, a cooperative transportation mechanism can adjust the suspending forces dynamically, ensuring a stable attitude for the suspended load. Different transportation structures are displayed in Table 1.

Table 1: Comparison of different transportation structures. Rigid attachment v.s. soft attachment

Type	Rigid	Soft
Connector	Gripper, Manipulator, Magnets	Cable
Advantages	Grasp ability, avoiding undesired oscillations, more actuated degrees of freedom	Low cost, lightweight, safety, fast deployment
Drawbacks	Difficult to design, more energy-consuming for active grippers, additional inertia for payload	Inability to exert negative (pushing) forces, fewer actuated degrees of freedom, undesired oscillations

Why No explicit communication A common prerequisite for collaborative robot transportation is a dependable, low-latency communication channel. This is not an issue for ground robots, as they are in close proximity and have the capacity to transport “heavy” high-performance communication equipment, which can enhance communication bandwidth, mitigate communication latency, and reduce failures (26). However, due to the limited payload capacity of aerial robots, they face challenges in establishing communication environments comparable to those of ground robots. As a result, performing trajectory planning independently on the onboard computing device of each aerial robot presents a viable solution (27–29). The cooperative transport strategy, which does not rely on explicit communication, is usually based on the leader-follower cooperative paradigm (30–32). Each aerial robot then follows its own planned trajectory. However, this approach places high demands on the homogeneity of the flying robots, as both the load distribution and trajectory planning primarily follow the principle of equal distribution. This is not easy to achieve in flying robots with different capabilities and limitations, particularly when the leader fails. A fixed

formation of robots is also a viable solution for cooperative transportation, where the problem of cooperative transport is transformed into a formation control problem (33–36). In this framework, the flying robots only need to maintain an appropriate distance from their neighbors to sustain the formation. Nevertheless, extending this method to an arbitrary number of flying robots is not straightforward, and the aforementioned approaches are not easily applicable to aerial robots with varying capabilities and limitations. The pre-designed formations of general solutions impede the flexibility and scalability of aerial robots, rendering them susceptible to failure during the procedure. Admittance control offers an alternative solution for cooperative transportation without the need for explicit communication (37, 38). By building upon the leader-follower framework, the follower robot estimates the external forces acting on it using only its model and the onboard inertial measurement unit (IMU). Based on this estimation, the robot then adjusts its dynamics to follow a reference trajectory, which is dynamically modified by the admittance controller according to both the estimated external force and the desired trajectory. This method is particularly applicable to a swarm of follower robots with varying capabilities and limitations, as well as in scenarios involving human-robot cooperation (39). The flexibility of this approach makes it suitable for a wide range of applications where robots need to respond to external forces while maintaining their collective mission objectives. However, it is important to note that admittance control requires a leader to guide the robot swarm, and it is inapplicable in situations where the cable is slack or relaxed. In such cases, the system’s stability and coordination could be compromised, limiting its effectiveness. Performance analysis of different methods is demonstrated in Table 2.

Table 2: Performance analysis of different methods. “+” represents strength/requirement level.

Method \ Performance	Scalability	Robustness	Communication Requirement	Fault Tolerance
Formation-based (36)	++	+	+	+
Payload Leader (40)	+	+	+	+
Admittance Control (38)	++	+	+	++
Dissipative Control	+++	+++	+	+++

Overall, in complex environments, maintaining explicit communication between the flying robots is not always feasible, leading to non-trivial instabilities and coordination issues (41). As the number of flying robots increases, the challenge of efficiently allocating and coordinating flying

robots becomes a significant hurdle during the cooperative transportation process. Additionally, halfway failures, where individual robots may experience malfunctions or unexpected behavior, further complicate the cooperative process. These factors—load allocation, communication limitations, halfway failures, and environmental complexities—are crucial considerations in the design of a scalable, robust, and communication-free transport framework for a swarm of flying robots. Addressing these challenges is key to ensuring reliable and efficient multi-robot cooperation in real-world applications.

What can be learned from tables To this end, we put forth a dissipative system as a flexible yet robust transport paradigm. The “table leg support problem”, as proposed by Euler in his 1774 dissertation at the St. Petersburg Academy of Sciences, aims to determine the four reaction forces of the legs of a rectangular table when a weight is placed on its surface (42). This problem is viewed as a redundant control problem, as a table supported by four legs has only three degrees of freedom—pitch, roll, and height—when considered as a static system. With four or more supporting table legs, the “smart” table automatically determines four or more reaction forces based on the load weight. Such cooperation structures are frequently observed in nature in Fig. 1A, and in all possible cases, the one that occurs is the one that uses the least energy. Euler proposed treating the supporting surface as a soft ground and introduced the concept of elastic deformation so that the solution for rigid support could be derived as a limiting case of the soft support scenario (15). As illustrated in Fig. 1B, table legs constitute a spring-damper structure. The more evenly the legs are distributed around the table’s center of gravity, the more stable the table becomes. Even in the event of a failure of one or more legs, this balanced configuration helps maintain stability and prevents the table from tipping over. Based on this fundamental understanding of the dissipative structure of tables, a dissipative transportation paradigm is developed that allows the flying robots to be self-organized for resilient load transportation.

In this paper, the cooperative transportation is constructed as a dissipative system with flying robots, virtual nodes, and a suspended load as in Fig. 1B. Dissipative force is generated among virtual nodes and between virtual nodes and flying robots, based on local information from their neighbors. As illustrated in Fig. 1C, we demonstrate a two-dimensional dissipative transport system. This framework enables adaptive load distribution and remains robust to both robot failure and dynamic

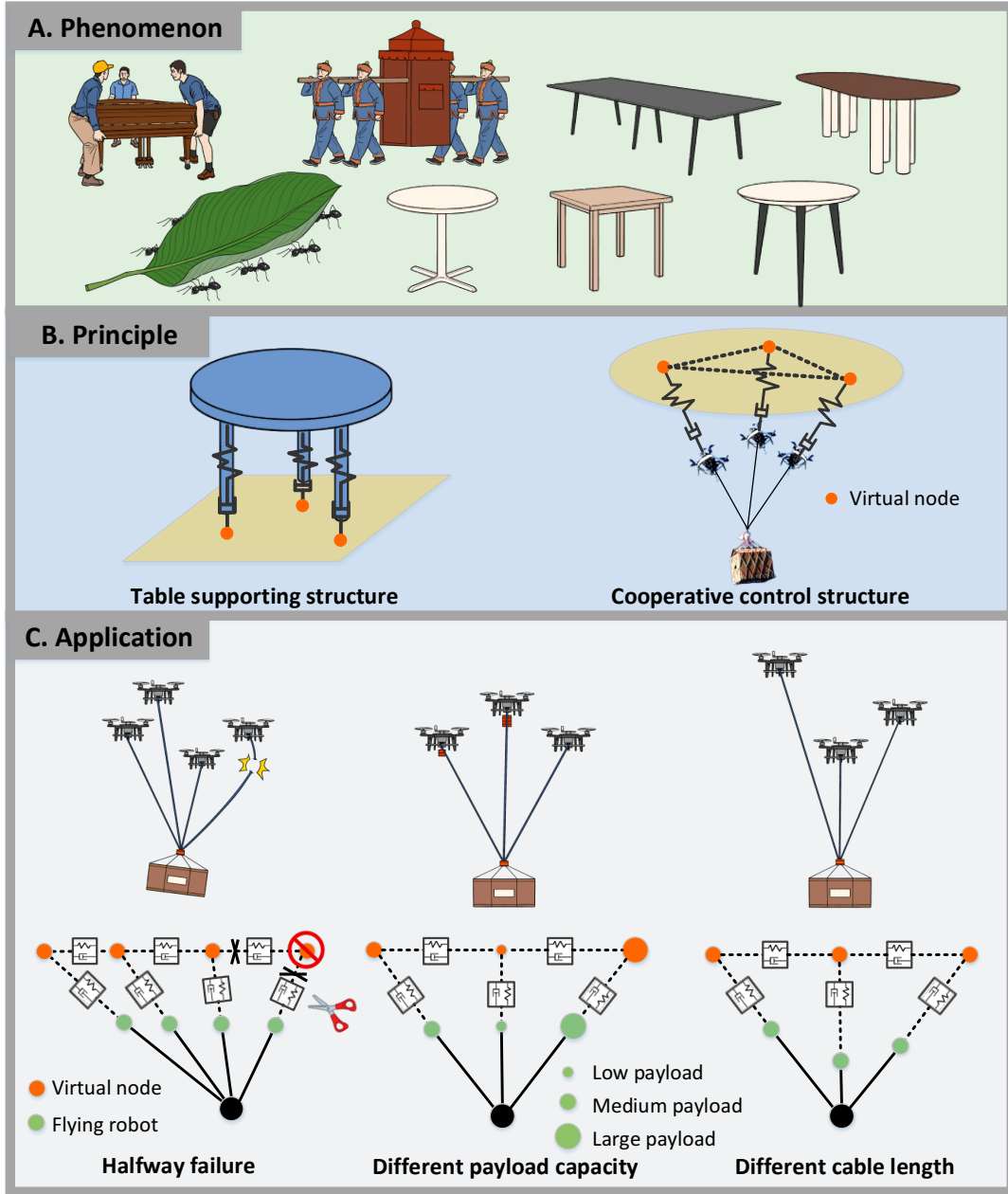


Fig. 1: Phenomenon, principle, and application of cooperative transportation. (A) Cooperative transport framework inspired by social and natural transport phenomena, and adaptive load distribution based on table structure. (B) Table supporting structure and a demonstration of three flying robots carrying a suspended load based on the design of virtual nodes. (C) Different cases of the dissipative cooperative transportation system. Green dots represent the flying robots, while orange dots denote their corresponding virtual nodes. The dotted lines illustrate the dissipation forces.

reconfiguration, such as the entry or exit of individual flying robots during transport. In the case of halfway failure, the formation autonomously reconfigures through local dissipative interactions among the remaining three flying robots, restoring equilibrium without external intervention. When a new robot joins during transport, the newly involved robot is connected to the payload and its neighbors, enabling the system to stabilize into a new equilibrium state. For heterogeneous payload distributions, the stability of the suspended load can be regulated by tuning the coefficient of virtual “spring”, and the dissipative interaction mechanism enhances system robustness by tolerating uncertainties in cable length.

Building on the above demonstration, this table-mechanics-inspired system flexibly adapts to variations in robot team size and maintains performance in the event of mid-flight failure. To the best of our knowledge, this is a collaborative transport framework that simultaneously addresses cable length uncertainty, communication discontinuity, potential robot failure, and supports heterogeneous robots in *the first time*, which are key challenges for real-world deployment.

Results

The proposed cooperative transportation system is validated through a combination of simulations and real-world flight experiments. In physical tests, we implement a cooperative aerial transport system composed of flying robots, suspension cables, hooks, and payloads. Extensive outdoor experiments are conducted to evaluate the proposed transportation system under diverse scenarios, including robustness to cable length uncertainties, resilience to heterogeneous payload capacities, and the ability to dynamically remove flying robots during transport.

Five Flying Robots Cooperative Transport In real-world flight experiments, we evaluate the transport system with teams of four, five, and six flying robots tasked with delivering a payload to a target location. A detailed analysis of the experiment with five flying robots is provided. Here, we focus on a representative scenario involving five aerial robots collaboratively transporting a 5 kg payload. After taking off, the load is taken to a height of 15 m above the ground and is precisely delivered to a second-floor terrace located approximately 34 m away and 5 m above the initial takeoff point. The experimental results are shown in Fig. 3. Fig. 3A illustrates the positions and

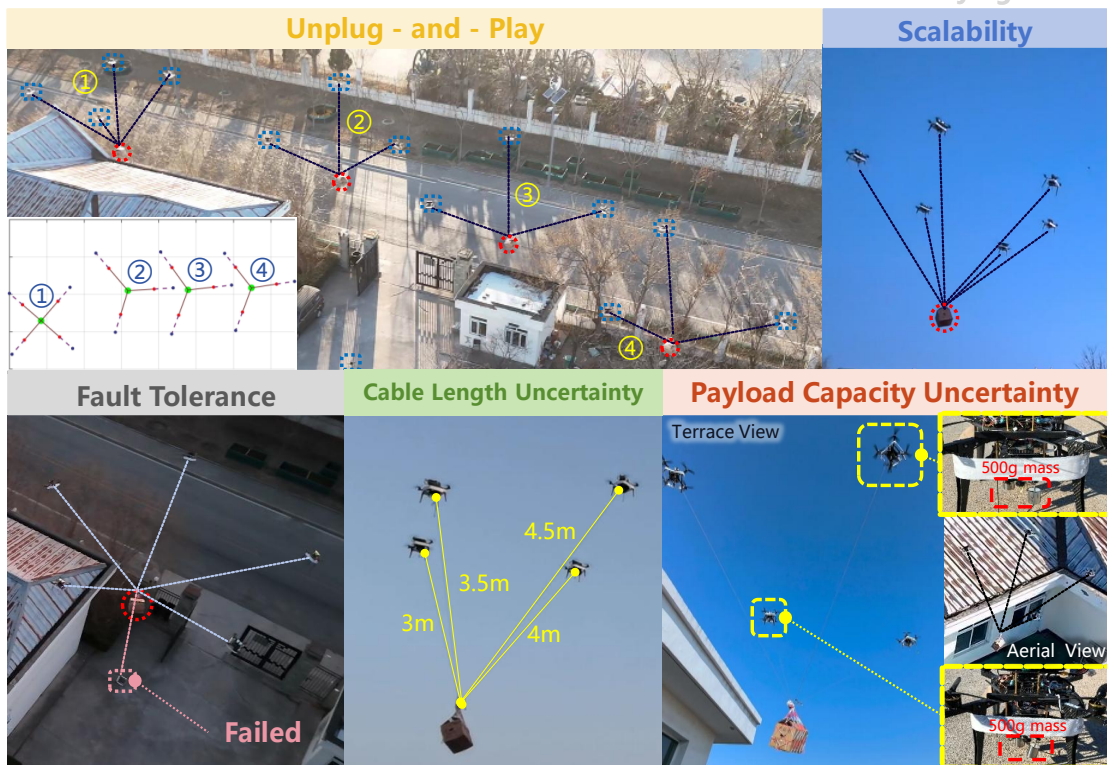
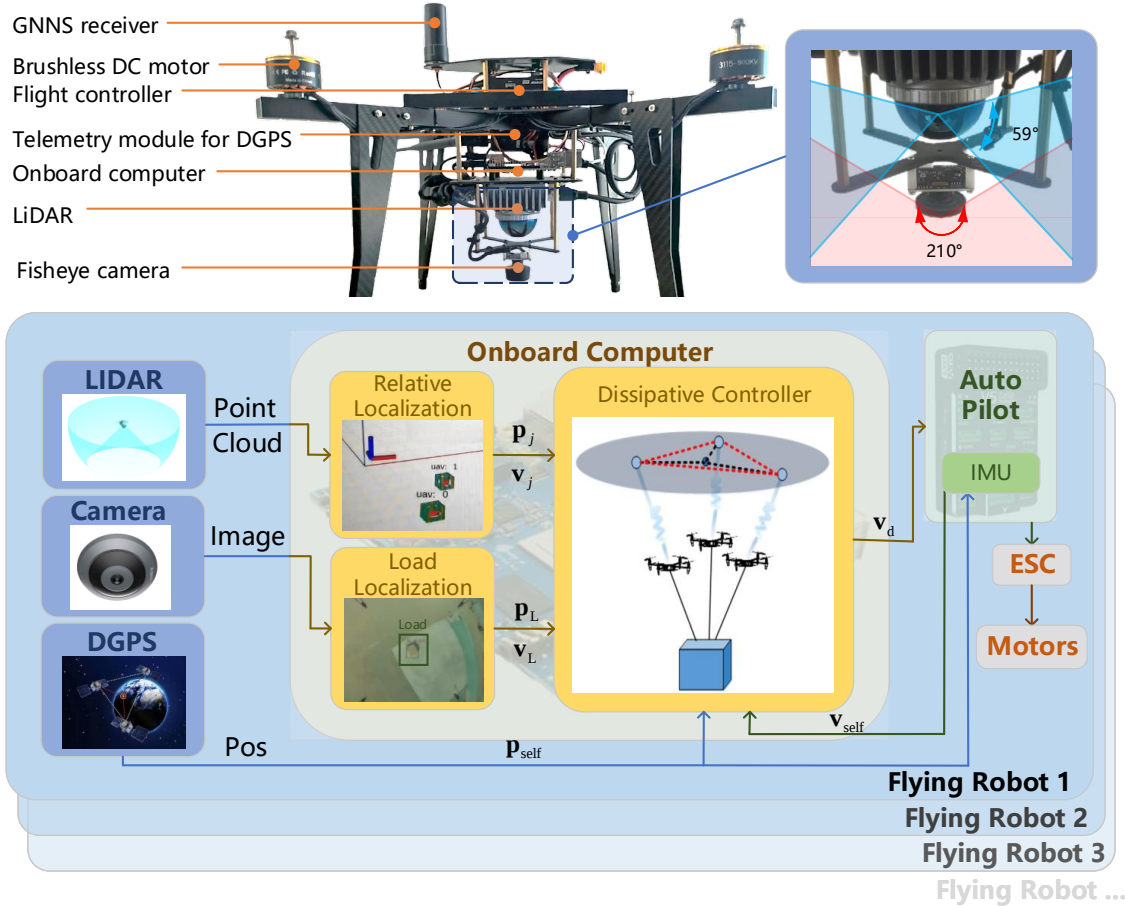


Fig. 2: Configuration and capabilities of flying robots. The hardware configuration, multi-robot framework, and transportation challenges are demonstrated.